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Writing a Paper

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Writing Process - 1

- Typical steps, but they may not necessarily be sequential
 - Pre-writing
 - Organizing
 - Writing
 - Proofreading
 - Revising

Writing Process - 2

1. Think about the topic you want to present, for some days or weeks.
2. Make figures and tables.
3. Then write as quickly as possible, as if thinking out loud. Get everything down, ignoring spelling, grammar, style and troublesome words.
4. Correct and rewrite only when the whole text is on paper.

Source: <https://www.elsevier.com/connect/writing-a-science-paper-some-dos-and-donts>

Writing Process - 3

- A paper is typically written by more than one author
- Different strategies can be applied
 - One: Do not split paper among co-authors
 - Better, main author writes first complete draft
 - Co-authors then add or change texts
 - Another: brainstorm the content, then each author writes a separate section
 - May lead to inconsistency in writing styles and flow

Source: <https://www.elsevier.com/connect/writing-a-science-paper-some-dos-and-donts>

Writing Process - 4

■ First draft

- ❑ Should begin as soon as the research is started
- ❑ Write as quickly as possible, as if thinking out loud. Get everything down, ignoring spelling, grammar, style and troublesome words
- ❑ If you are too careful on the first draft, you may not be able to write anything at all
- ❑ When you are stuck, just write anything, regardless how bad it is
- ❑ Writing simple statements may help

Sources: 1. <https://www.elsevier.com/connect/writing-a-science-paper-some-dos-and-donts>
2. Justin Zobel, "Writing for Computer Science" (2nd Edition), Springer.

Writing Process - 5

- From first draft to submission
 - ❑ Brainstorm
 - ❑ Prepare outline with section titles
 - ❑ Write the introduction section
 - ❑ Write body
 - ❑ Write related work
 - ❑ Write conclusion
 - ❑ Write abstract
 - ❑ Revise

Sources: Justin Zobel, “Writing for Computer Science” (2nd Edition), Springer.

Pre-writing - 1

- Elements to think about
 - ❑ Topic, audience, genre, style, opportunity, research, the writer, and purpose
- Ask about your audience
 - ❑ Who is the audience?
 - ❑ Is your audience interested in the topic?
 - ❑ Why is your topic interesting to them?
 - ❑ Why should your audience read your writing?
 - ❑ What does your audience already know and will know about this topic?
 - ❑ What do you hope your audience to know about the topic?

Pre-writing - 2

- When starting to write, you may not be sure what to say. It's normal.
- Strategies to get started
 - Explore the problem: reader? purpose? who are you?
 - Brainstorm
 - Keep writing, don't censor or evaluate, keep returning to the problem
 - Talk to your reader
 - Ask yourself questions
 - Who, what, where, when, how?

Source: https://owl.purdue.edu/owl/general_writing/the_writing_process/prewriting/questions.html

Pre-writing - 3

- Scope of the paper
 - Scope addresses the purpose or aim of the paper
 - What to include, what might be included
 - Questions may help identify the scope
 - Which results are the most surprising?
 - What is the one result that other researchers might adopt in their work?
 - Are the other outcomes independent enough to be published separately later on?
 - Are they interesting enough to justify their being included?

Source: Justin Zobel, "Writing for Computer Science" (2nd Edition), Springer

The idea

- Idea: a re-usable insight, useful to the reader
- Your paper should have just one “ping”: **one clear, sharp idea**
- You may not know exactly what the ping is when you start writing; but you must know when you finish
- If you have lots of ideas, write lots of papers

Source: “[How to write a great research paper - Seven simple suggestions](http://research.microsoft.com/en-us/um/people/simonpj/papers/giving-a-talk/giving-a-talk.htm)” by

Simon Peyton Jones, Microsoft Research, Cambridge

<http://research.microsoft.com/en-us/um/people/simonpj/papers/giving-a-talk/giving-a-talk.htm>

Can you hear the “ping”?

- Many papers contain good ideas, but do not distill what they are.
- Make certain that the reader is in no doubt what the idea is. Be 100% explicit:
 - “The main idea of this paper is....”
 - “In this section we present the main contributions of the paper.”

Source: “[How to write a great research paper - Seven simple suggestions](http://research.microsoft.com/en-us/um/people/simonpj/papers/giving-a-talk/giving-a-talk.htm)” by

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Tell a story

Imagine you are explaining at a whiteboard

- Here is a problem
- It's an interesting problem
- It's an unsolved problem
- Here is my idea
- My idea works (details, data)
- Here's how my idea compares to other people's approaches

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Exercise 1

- Work in groups of 4-5 each, discuss and answer the following questions, based on your experience in reading research papers
 - What elements constitute a research paper? Are these elements appearing all papers? If not, which elements are most commonly found? Are they follow the same order in all papers?
 - What do each element have in common?
 - Are there differences between conference and journal papers?
- Discuss in 15 minutes and present to the whole class in 10 minutes

Organizing (conference paper)

- Title (1000 readers)
- Abstract (4 sentences, 100 readers)
- Introduction (1 page, 100 readers)
- The problem (1 page, 10 readers)
- My idea (2 pages, 10 readers)
- The details (5 pages, 3 readers)
- Related work (1-2 pages, 10 readers)
- Conclusions and further work (0.5 pages)

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Title and author

- Title is very important for a paper
 - It is the first thing people notice about your paper
 - It is the title that people decide to read your paper
- Title needs to be concise, describing content sufficiently and conveying the idea
- It contains keywords, usually at the beginning
- If possible, name your method in the title
 - “SRD: a smart reader of human emotions”
- Author: use your name consistently across all papers

Abstract - 1

- Single paragraph of about 50-200 words, around 5 sentences
- Help readers to decide whether the paper is relevant to them
- Abstract structure
 - Short motivation (problem), Proposed solution, Evaluation, Evaluation results
 - Some journals require to have: context, objectives, method, results, and conclusions

Abstract - 2

■ Abstract writing pitfalls

- ❑ Don't put unexplained or undefined terms whose meanings are not well known
- ❑ Solutions: explain them; rephrase them using plain words; not get into too much detail (without mentioning them).
- ❑ Minor details or a description of the structure of the paper are usually inappropriate
 - Acronyms, mathematics, abbreviations, or citations
 - Rare: if citations are needed, provide full citations.
 - ❑ Ex. Don't put [1] in the abstract

Introduction - 1

- An introduction can be regarded as an expanded version of the abstract.
- It describes the paper's topic, the problem being studied, references to key papers, the approach to the solution, the scope and limitations of the solution, and the outcomes.
- There needs to be enough detail to allow readers to decide whether or not they need to read further.

Introduction - 2

■ Stirewalt's 5-paragraph rule

□ Introductory paragraph

- What is the problem?
- Why is it relevant?
- Why is the problem hard?
- What is your solution?

□ Background paragraph

- Elaborate on why is the problem is hard
- Examining prior work, point out one or two shortcomings that your solution overcomes

□ Transition paragraph

- What do you apply to overcome shortcomings

□ Details paragraph

- Technical challenges do you have to overcome
- What kinds of evaluation you perform

□ Assessment paragraph

- Assess your results and briefly state the broadly interesting conclusions that these results support.

Source: <http://www.cse.msu.edu/%7Echengb/Writing/intro-guidelines-stirewalt.txt>

Body

- Details the problem, my ideas, and the details
- Typically consisting of
 - Background – knowledge to understand the method
 - Proposed Method – method proposed by paper
 - Experiment – details of experiment to validate the method
 - Results – what are obtained from the experiment, focusing on reporting and describing results
 - Discussions – interpreting the results, findings, implications, comparisons with previous work
- Possible variations dependent on conferences and journals

Related work/Literature review - 1

- Different from background, related work discusses alternative approaches to the same problem addressed in your paper
 - Different section from background
- Compare and contrast existing approaches, related them to your approach
- Provide strengths and weaknesses of related work

Related work/Literature review - 2

- Where to put the related work section
 - After the introduction/example section
 - Before the conclusion section
- After the introduction/example section
 - Pros: immediately clear out reviewers' wonder on how the work differs from previous work
 - Cons: hard to let readers to know what you are talking about before showing the approach details
- Before the conclusion section
 - Pros: Now reviewers' know what your approach is about
 - Cons: reviewers keep wondering how the work differs from previous work till this point
 - But for very closely related work, you should have pointed out the differences in the introduction section

Conclusion (and future work)

- Often easy to write conclusions
 - Nothing here should surprise readers; simply summarize your contributions and findings
 - In the introduction, “We propose a new approach ...” vs. In the conclusions, “We have proposed a new approach ...”
 - Restate key points: method, results, and contributions
- You can state the broader impacts of your approach
- You can optionally describe limitations and future work here if you don’t have a discussion section for them and propose future work

Exercise 2

- Work with your potential partner of term project
- Read one of the papers from <https://drive.google.com/drive/folders/17F2MutsUiq72FC3HiqX6rF7yZpwsHExu?usp=sharing>
- OR choose a paper you read today
- Read the introduction section
- Write an abstract for the paper in 100 - 200 words
 - Condition: the abstract must differ from the original one